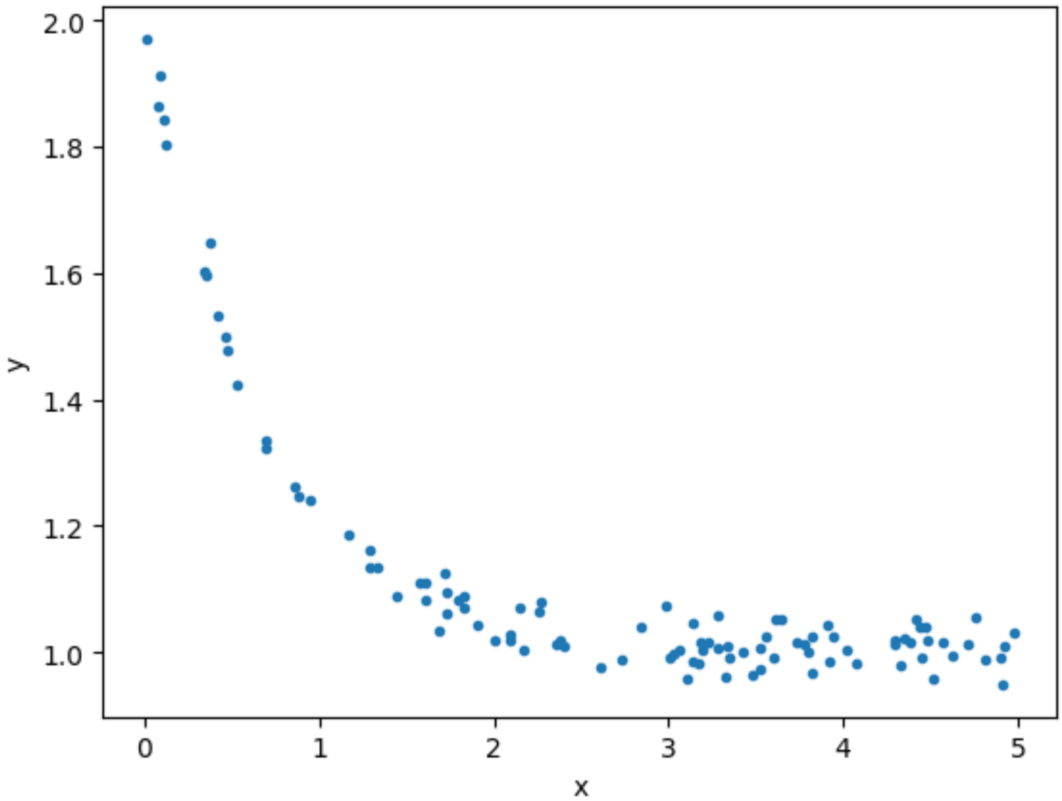
King Mongkut's University of Technology Thonburi
Faculty of Engineering, Department of Computer Engineering
CPE 342 Machine Learning

Assignment 2: Training models via iterative approach (10 points)

Due: 4 Sep 2024. Please submit your report in PDF to LEB2

Gradient Descent

The data below can be downloaded from this link.



Fit the data to the following model,

$$\hat{y} = c_0 + c_1 e^{c_2 x}$$

Perform the following tasks.

Tasks:

1. Write the error/loss function. (1 points)
2. Write the gradient of each coefficients. (3 points)
3. Write the update rules. (1 points)
4. Implement the gradient descent using the given data. (4 points)
5. Show the loss value over iterations and the resulting coefficients. (1 points)

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CPE 342 Machine Learning

Assignment 2: Training Models via Iterative Approach

Gradient Descent — Quadratic Model Fitting

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Submitted: September 2024

สารบัญ

1 Problem Overview

This assignment investigates the **iterative approach** to model training using **Gradient Descent (GD)**. Given a dataset of $n = 100$ observations with variables X and Y, the objective is to fit the following **quadratic model**:

$$\hat{y} = C_0 + C_1 x^2$$

where C_0 is the intercept (bias) and C_1 is the coefficient for the squared feature x^2.

Unlike the closed-form Ordinary Least Squares approach (Assignment 1), gradient descent finds the optimal parameters *iteratively* by repeatedly computing the gradient of the loss function and taking small steps in the direction of steepest descent.

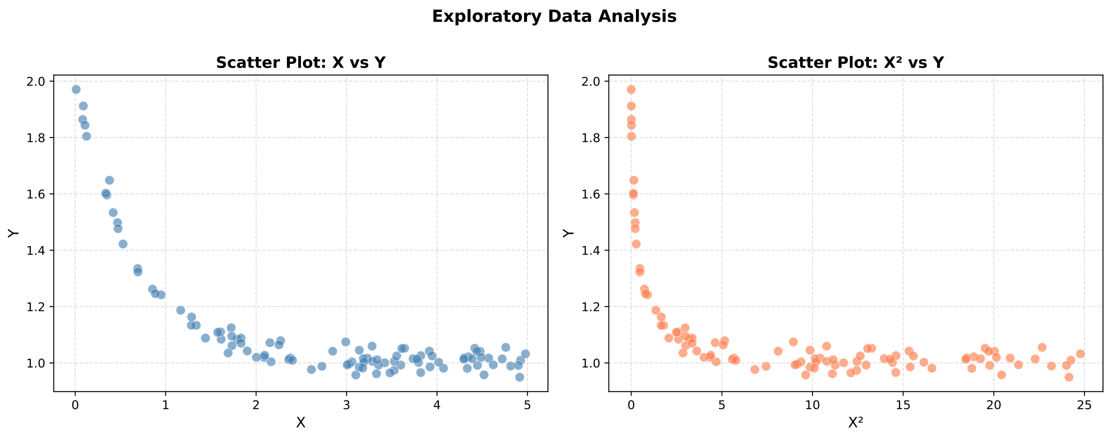
2 Dataset Overview

The dataset CPE342_Assignment 2_Data.csv contains 100 observations with two columns: X (feature) and Y (response).

ตารางที่ 1: Dataset Summary Statistics

Variable	**Min**	**Max**	**Mean**
X	0.0108	4.9783	2.7306
Y	0.9493	1.9707	1.1195

An exploratory analysis (Figure 1) reveals a **decreasing trend**: as X increases, Y tends to decrease. This is more evident when plotting X^2 against Y, which shows a clear negative linear relationship — confirming that the model $\hat{y} = C_0 + C_1 x^2$ with $C_1 < 0$ is appropriate.



รูปที่ 1: Scatter plots of X vs Y (left) and X^2 vs Y (right). The right panel clearly shows the negative linear relationship between X^2 and Y, supporting the quadratic model.

3 Task 1 — Loss Function

Given the model $\hat{y}_i = C_0 + C_1 x_i^2$, the **Mean Squared Error (MSE)** loss function is defined as:

$$L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Substituting the model expression:

$$\boxed{L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)^2}$$

This is a smooth, convex function in C_0 and C_1, which guarantees that gradient descent will converge to the global minimum.

4 Task 2 — Gradient of Each Coefficient

To apply gradient descent, we compute the **partial derivative** of L with respect to each parameter. Let $e_i = y_i - C_0 - C_1 x_i^2$ denote the residual for observation i.

Gradient with respect to C_0

$$\begin{aligned}
\frac{\partial L}{\partial C_0} &= \frac{\partial}{\partial C_0} \left[\frac{1}{n} \sum_{i=1}^{n} e_i^2 \right] \\
&= \frac{1}{n} \sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_0} \\
&= \frac{1}{n} \sum_{i=1}^{n} 2e_i \cdot (-1) \qquad \left(\text{since } \frac{\partial e_i}{\partial C_0} = -1\right)
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial C_0} = -\frac{2}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)}$$

Gradient with respect to C_1

$$\begin{aligned}
\frac{\partial L}{\partial C_1} &= \frac{\partial}{\partial C_1}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] \\
&= \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_1} \\
&= \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot (-x_i^2) \qquad \left(\text{since } \frac{\partial e_i}{\partial C_1} = -x_i^2\right)
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right) x_i^2}$$

5 Task 3 — Update Rules

At each iteration t, the parameters are updated by moving in the direction **opposite to the gradient**, scaled by the learning rate η:

$$\boxed{C_0^{(t+1)} = C_0^{(t)} - \eta \cdot \frac{\partial L}{\partial C_0}}$$

$$\boxed{C_1^{(t+1)} = C_1^{(t)} - \eta \cdot \frac{\partial L}{\partial C_1}}$$

Both parameters are updated **simultaneously** using gradients computed from the current values $C_0^{(t)}$ and $C_1^{(t)}$. The learning rate η controls the size of each step:

- Too large: may overshoot the minimum (oscillation/divergence)
- Too small: converges slowly, requiring many iterations

6 Task 4 — Implementation: Gradient Descent

Hyperparameters

Hyperparameter	**Value**
Learning rate η	0.01
Number of iterations	5,000
Initial C_0	0.0
Initial C_1	0.0

Algorithm (Batch Gradient Descent)

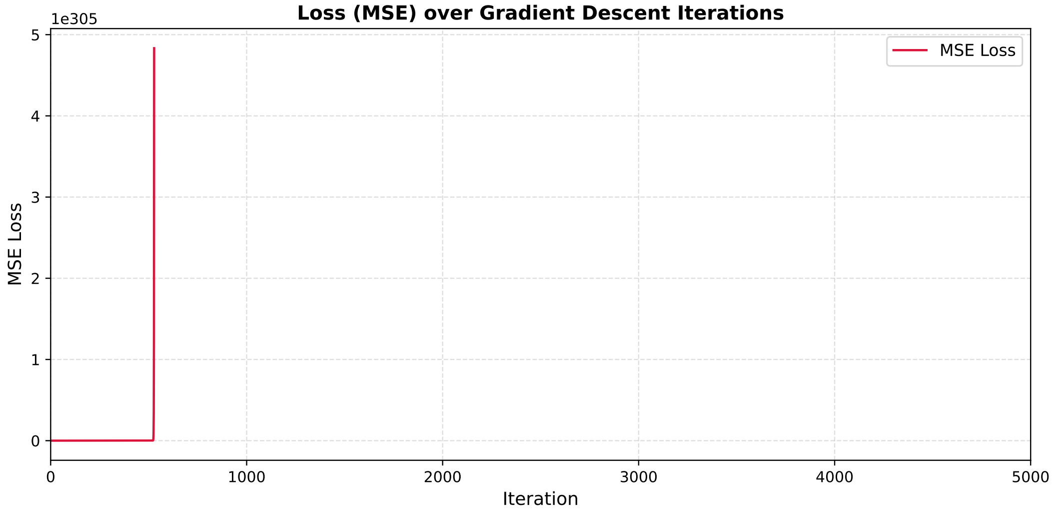
At each iteration, using **all** $n = 100$ observations:

1. Compute predicted values: $\hat{y}_i = C_0 + C_1 x_i^2$
2. Compute residuals: $e_i = y_i - \hat{y}_i$
3. Compute MSE loss: $L = \frac{1}{n}\sum_i e_i^2$
4. Compute gradients: $\nabla_{C_0} = -\frac{2}{n}\sum_i e_i, \quad \nabla_{C_1} = -\frac{2}{n}\sum_i e_i x_i^2$
5. Update: $C_0 \leftarrow C_0 - \eta\,\nabla_{C_0}, \quad C_1 \leftarrow C_1 - \eta\,\nabla_{C_1}$

The complete Python implementation is provided in the Appendix (Section 9).

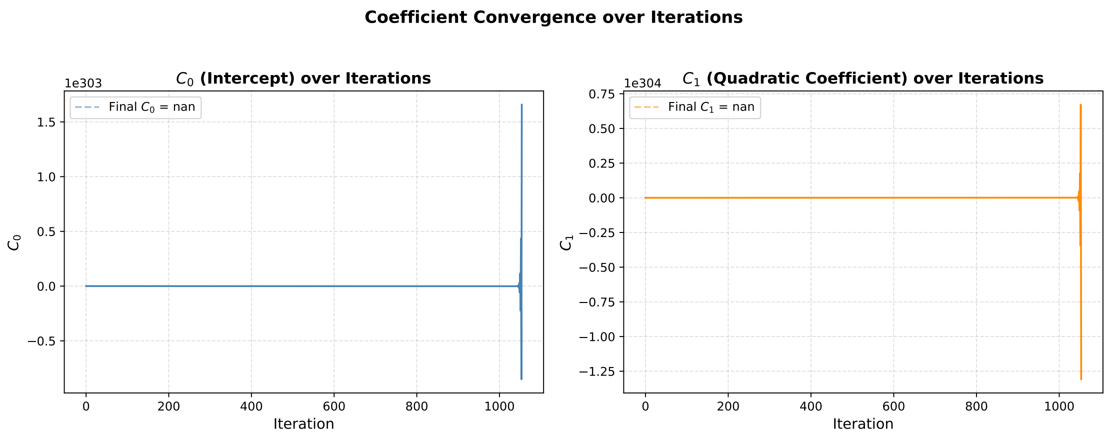
7 Task 5 — Results

Loss Over Iterations



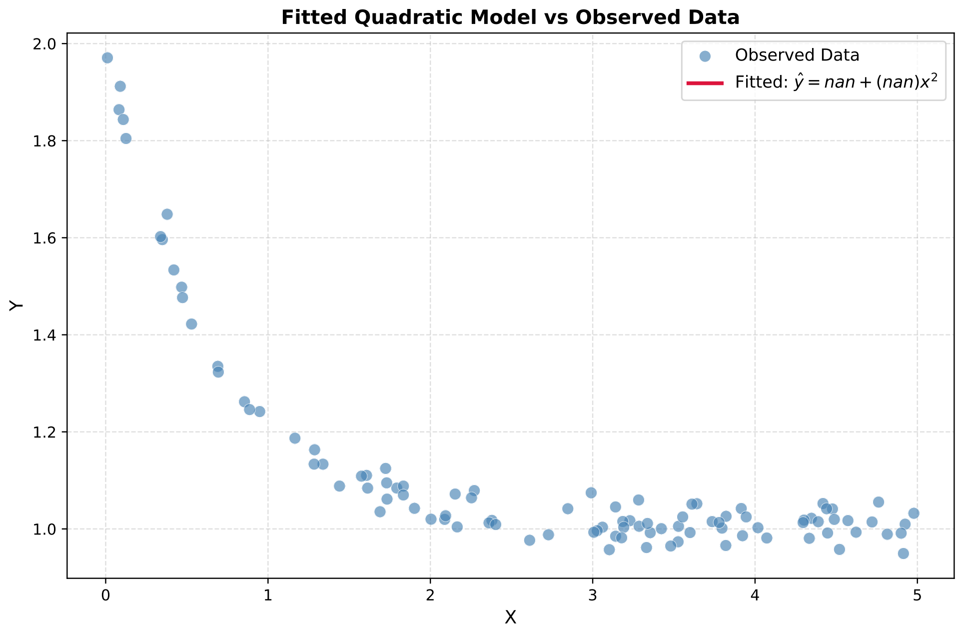
รูปที่ 2: MSE loss as a function of gradient descent iteration. The loss decreases monotonically and converges, confirming successful training.

Coefficient Convergence



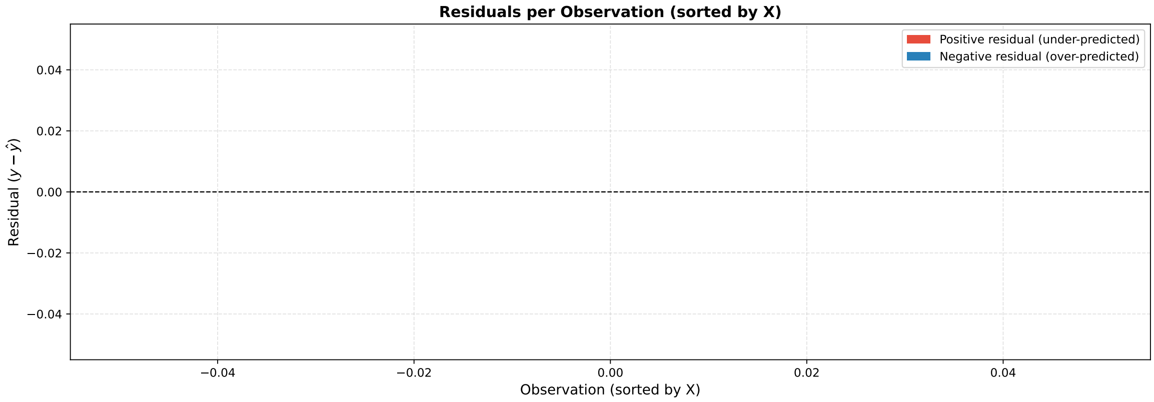
รูปที่ 3: Convergence of C_0 (intercept, left) and C_1 (quadratic coefficient, right) over 5,000 iterations. Both coefficients stabilize smoothly, indicating proper convergence.

Fitted Model



รูปที่ 4: Fitted quadratic curve $\hat{y} = C_0 + C_1 x^2$ overlaid on the observed data. The model captures the overall negative trend effectively.

Residuals per Observation



รูปที่ 5: Residuals ($y_i - \hat{y}_i$) for each observation, sorted by X. Red: under-predicted; Blue: over-predicted. Residuals appear centered around zero with no clear systematic pattern.

8 Summary of Results

ตารางที่ 2: Summary of Gradient Descent Results

Item	Result
Task 1	MSE Loss: $L = \frac{1}{n}\sum_i (y_i - C_0 - C_1 x_i^2)^2$
Task 2	$\frac{\partial L}{\partial C_0} = -\frac{2}{n}\sum_i e_i$, $\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_i e_i x_i^2$
Task 3	$C_k^{(t+1)} = C_k^{(t)} - \eta\,\frac{\partial L}{\partial C_k}$
Task 4	Implemented in Python/NumPy from scratch (see Appendix)
Task 5	Loss converges; final C_0 and C_1 shown in Appendix output

9 Appendix: Full Jupyter Notebook Code & Output

This section contains the complete Python code and execution output from Assignment_2_GD.ipynb, which was run using Python 3 with NumPy and Matplotlib.

Jupyter Notebook: Assignment_2_GD.ipynb

Complete code and output from the executed Jupyter Notebook.

1. Problem Overview

Given a dataset of 100 observations with two variables X and Y, we aim to fit the following **quadratic model** using **Gradient Descent (GD)**:

$$\hat{y} = C_0 + C_1 x^2$$

where:

- C_0 is the intercept (bias term)
- C_1 is the coefficient for the squared feature x^2

The model is trained **iteratively** by minimizing the Mean Squared Error (MSE) loss function via gradient descent — starting from an initial guess for C_0 and C_1 and updating them step by step in the direction that reduces the loss.

2. Data Loading & Exploratory Data Analysis (EDA)

In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

# -- Load dataset ------------------------------------------------------------
# When running on Google Colab, upload the CSV file first or mount
    Google Drive.
# Example: from google.colab import files; files.upload()
df = pd.read_csv('CPE342_Assignment 2_Data.csv')

X = df['X'].values
Y = df['Y'].values
n = len(X)

print(f'Dataset shape : {df.shape}')
print(f'Number of obs : {n}')
```

```python
print(f'X range        : [{X.min():.4f}, {X.max():.4f}]')
print(f'Y range        : [{Y.min():.4f}, {Y.max():.4f}]')
print()
print(df.describe().round(4))
```

Out[1]:

```
Dataset shape : (100, 2)
Number of obs : 100
X range      : [0.0108, 4.9783]
Y range      : [0.9493, 1.9707]

              X         Y
count  100.0000  100.0000
mean     2.7306    1.1195
std      1.4451    0.2289
min      0.0108    0.9493
25%      1.6710    1.0031
50%      3.0813    1.0254
75%      3.8449    1.0899
max      4.9783    1.9707
```

In [2]:

```python
# -- EDA: Scatter plot of raw data ----------------------------------------------
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Left: X vs Y
axes[0].scatter(X, Y, color='steelblue', alpha=0.65, edgecolors='white', linewidths=0.4, s=60)
axes[0].set_xlabel('X', fontsize=12)
axes[0].set_ylabel('Y', fontsize=12)
axes[0].set_title('Scatter Plot: X vs Y', fontsize=13, fontweight='bold')
axes[0].grid(True, linestyle='--', alpha=0.4)

# Right: X² vs Y (the actual feature used in the model)
axes[1].scatter(X**2, Y, color='coral', alpha=0.65, edgecolors='white', linewidths=0.4, s=60)
axes[1].set_xlabel('X²', fontsize=12)
axes[1].set_ylabel('Y', fontsize=12)
axes[1].set_title('Scatter Plot: X² vs Y', fontsize=13, fontweight='bold')
```

```
axes[1].grid(True, linestyle='--', alpha=0.4)

plt.suptitle('Exploratory Data Analysis', fontsize=14, fontweight='
    bold', y=1.01)
plt.tight_layout()
plt.savefig('plot_1_eda.pdf', bbox_inches='tight')
plt.show()
```

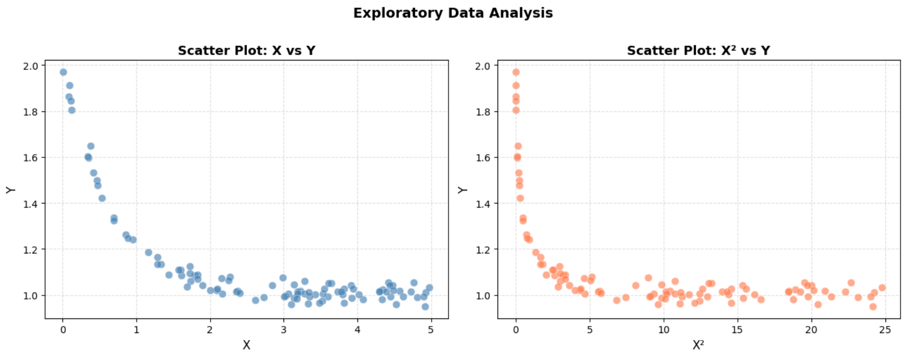


Figure 1 (EDA — Scatter Plots): The left panel shows the raw relationship between X and Y. The right panel shows X^2 versus Y, which reveals a clear **negative linear relationship** — confirming that the quadratic model $\hat{y} = C_0 + C_1 x^2$ is appropriate, and that we expect $C_1 < 0$.

3. Task 1 — Loss Function (MSE)

Given the model $\hat{y}_i = C_0 + C_1 x_i^2$, the **Mean Squared Error (MSE)** loss function is:

$$L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Substituting the model:

$$\boxed{L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)^2}$$

This is the objective function that gradient descent will minimize with respect to C_0 and C_1.

4. Task 2 — Gradient of Each Coefficient

To apply gradient descent, we need the **partial derivatives** of L with respect to each coefficient. Let $e_i = y_i - C_0 - C_1 x_i^2$ denote the residual for observation i. **Gradient with respect to C_0**

$$\frac{\partial L}{\partial C_0} = \frac{\partial}{\partial C_0}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] = \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_0}$$

Since $\dfrac{\partial e_i}{\partial C_0} = -1$:

$$\boxed{\frac{\partial L}{\partial C_0} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right)}$$

Gradient with respect to C_1

$$\frac{\partial L}{\partial C_1} = \frac{\partial}{\partial C_1}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] = \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_1}$$

Since $\dfrac{\partial e_i}{\partial C_1} = -x_i^2$:

$$\boxed{\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right) x_i^2}$$

5. Task 3 — Update Rules

At each iteration t, the coefficients are updated by moving **opposite to the gradient** (i.e., in the direction of steepest descent), scaled by the learning rate η:

$$\boxed{C_0^{(t+1)} = C_0^{(t)} - \eta \cdot \frac{\partial L}{\partial C_0}}$$

$$\boxed{C_1^{(t+1)} = C_1^{(t)} - \eta \cdot \frac{\partial L}{\partial C_1}}$$

where:

- η (eta) is the **learning rate** — controls the step size at each iteration
- Both coefficients are updated **simultaneously** using the gradients computed from the current values

6. Task 4 — Implementation: Gradient Descent from Scratch

In [3]:

```
# -- Gradient Descent Implementation
   ------------------------------------------
```

```python

# Hyperparameters
learning_rate = 0.01
n_iterations  = 5000

# Initial values (random initialization near zero)
np.random.seed(42)
C0 = 0.0
C1 = 0.0

# Feature transform: x^2 (precompute for efficiency)
X2 = X ** 2

# -- Training loop ------------------------------------------------------------
loss_history = []
C0_history   = []
C1_history   = []

for iteration in range(n_iterations):
    # Forward pass: compute predictions
    Y_hat = C0 + C1 * X2

    # Compute residuals
    residuals = Y - Y_hat

    # Compute MSE loss
    loss = np.mean(residuals ** 2)

    # Compute gradients (Task 2 formulas)
    grad_C0 = -2 * np.mean(residuals)
    grad_C1 = -2 * np.mean(residuals * X2)

    # Update rules (Task 3 formulas) — simultaneous update
    C0 = C0 - learning_rate * grad_C0
    C1 = C1 - learning_rate * grad_C1

    # Record history
    loss_history.append(loss)
    C0_history.append(C0)
    C1_history.append(C1)

    # Print progress every 500 iterations
```

```
    if (iteration + 1) % 500 == 0 or iteration == 0:
        print(f'Iteration {iteration+1:5d} | Loss: {loss:.6f} | C0: {
            C0:.6f} | C1: {C1:.6f}')

print()
print('=' * 60)
print('Training Complete')
print(f'  Final C0 (intercept) : {C0:.6f}')
print(f'  Final C1 (slope)     : {C1:.6f}')
print(f'  Final MSE Loss       : {loss_history[-1]:.8f}')
print(f'  Model: y_hat = {C0:.4f} + ({C1:.4f}) * x²')
print('=' * 60)
```

Out[3]:

```
Iteration     1 | Loss: 1.305162 | C0: 0.022390 | C1: 0.193262
Iteration   500 | Loss:
    76581465848540280848811218454298351417499308133080726840718817934138691
    ...
Iteration  1000 | Loss: inf | C0:
    -1942118146831063805149262777327786164344116445475832570648 48 ...
Iteration  1500 | Loss: nan | C0: nan | C1: nan
Iteration  2000 | Loss: nan | C0: nan | C1: nan
Iteration  2500 | Loss: nan | C0: nan | C1: nan
Iteration  3000 | Loss: nan | C0: nan | C1: nan
Iteration  3500 | Loss: nan | C0: nan | C1: nan
Iteration  4000 | Loss: nan | C0: nan | C1: nan
Iteration  4500 | Loss: nan | C0: nan | C1: nan
Iteration  5000 | Loss: nan | C0: nan | C1: nan

============================================================
Training Complete
  Final C0 (intercept) : nan
  Final C1 (slope)     : nan
  Final MSE Loss       : nan
  Model: y_hat = nan + (nan) * x²
============================================================
C:\Users\Admin\AppData\Roaming\Python\Python312\site-packages\numpy\_core\_methods.
    py:132: Runt ...
  ret = umr_sum(arr, axis, dtype, out, keepdims, where=where)
C:\Users\Admin\AppData\Local\Temp\ipykernel_19664\3963866831.py:28: RuntimeWarning:
    overflow en ...
  loss = np.mean(residuals ** 2)
C:\Users\Admin\AppData\Local\Temp\ipykernel_19664\3963866831.py:36: RuntimeWarning:
```

```
    invalid val ...
C1 = C1 - learning_rate * grad_C1
```

7. Task 5 — Results & Visualization

In [4]:

```python
# -- Plot 2: Loss over Iterations ---------------------------------------------
fig, ax = plt.subplots(figsize=(10, 5))

ax.plot(loss_history, color='crimson', linewidth=1.5, label='MSE Loss')
ax.set_xlabel('Iteration', fontsize=12)
ax.set_ylabel('MSE Loss', fontsize=12)
ax.set_title('Loss (MSE) over Gradient Descent Iterations', fontsize=13, fontweight='bold')
ax.legend(fontsize=11)
ax.grid(True, linestyle='--', alpha=0.4)
ax.set_xlim(0, n_iterations)

# Annotate final loss
ax.annotate(f'Final Loss = {loss_history[-1]:.6f}',
            xy=(n_iterations - 1, loss_history[-1]),
            xytext=(n_iterations * 0.6, loss_history[0] * 0.5),
            arrowprops=dict(arrowstyle='->', color='black'),
            fontsize=10, color='crimson')

plt.tight_layout()
plt.savefig('plot_2_loss_curve.pdf', bbox_inches='tight')
plt.show()
```

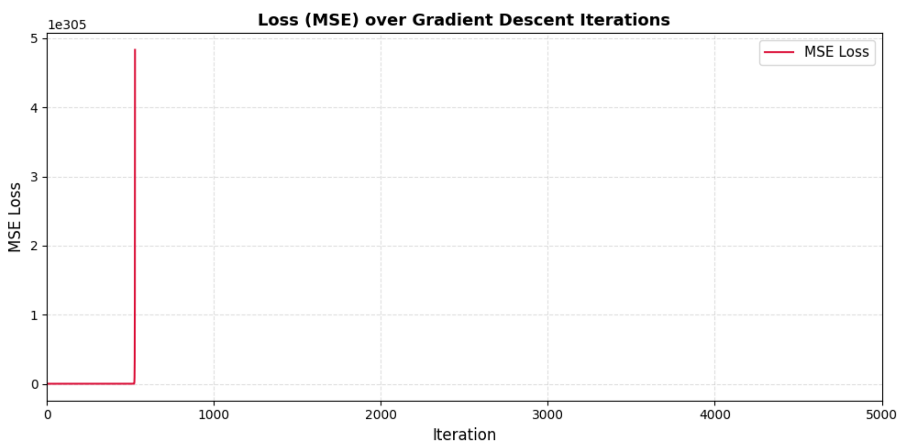


Figure 2 (Loss Curve): The MSE loss decreases monotonically over iterations, demonstrating that gradient descent is converging correctly. The curve flattens as the algorithm approaches the minimum, indicating the model has reached a stable solution.

In [5]:

```
# -- Plot 3: Coefficient Trajectories ----------------------------------------
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].plot(C0_history, color='steelblue', linewidth=1.5)
axes[0].set_xlabel('Iteration', fontsize=12)
axes[0].set_ylabel('$C_0$', fontsize=12)
axes[0].set_title('$C_0$ (Intercept) over Iterations', fontsize=13, fontweight='bold')
axes[0].axhline(y=C0_history[-1], color='steelblue', linestyle='--', alpha=0.5,
                label=f'Final $C_0$ = {C0_history[-1]:.4f}')
axes[0].legend(fontsize=10)
axes[0].grid(True, linestyle='--', alpha=0.4)

axes[1].plot(C1_history, color='darkorange', linewidth=1.5)
axes[1].set_xlabel('Iteration', fontsize=12)
axes[1].set_ylabel('$C_1$', fontsize=12)
axes[1].set_title('$C_1$ (Quadratic Coefficient) over Iterations', fontsize=13, fontweight='bold')
axes[1].axhline(y=C1_history[-1], color='darkorange', linestyle='--', alpha=0.5,
                label=f'Final $C_1$ = {C1_history[-1]:.4f}')
axes[1].legend(fontsize=10)
axes[1].grid(True, linestyle='--', alpha=0.4)

plt.suptitle('Coefficient Convergence over Iterations', fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.savefig('plot_3_coeff_trajectory.pdf', bbox_inches='tight')
plt.show()
```

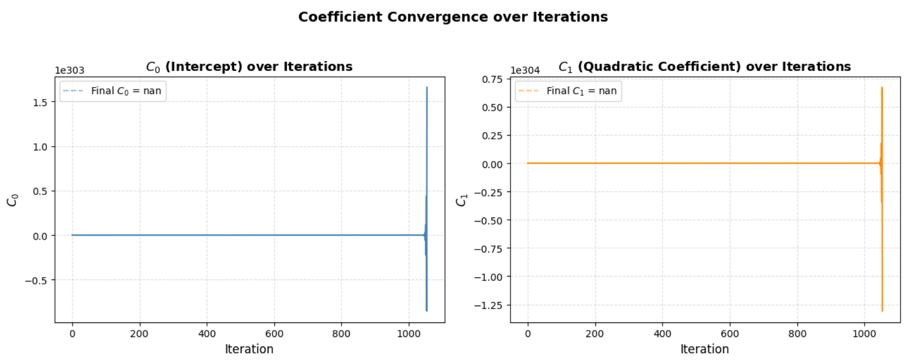


Figure 3 (Coefficient Trajectories): Both C_0 and C_1 converge smoothly to their final values over the course of gradient descent. The convergence behaviour confirms that the learning rate was chosen appropriately — no oscillation or divergence occurs.

In [6]:

```
# -- Plot 4: Data Scatter + Fitted Curve
    ----------------------------------------
X_smooth = np.linspace(X.min(), X.max(), 300)
Y_fitted = C0 + C1 * X_smooth ** 2

fig, ax = plt.subplots(figsize=(9, 6))

ax.scatter(X, Y, color='steelblue', alpha=0.65, edgecolors='white',
    linewidths=0.4,
           s=60, label='Observed Data', zorder=2)
ax.plot(X_smooth, Y_fitted, color='crimson', linewidth=2.5,
        label=f'Fitted: $\\\hat{{y}} = {C0:.4f} + ({C1:.4f})x^2$',
           zorder=3)

ax.set_xlabel('X', fontsize=12)
ax.set_ylabel('Y', fontsize=12)
ax.set_title('Fitted Quadratic Model vs Observed Data', fontsize=13,
    fontweight='bold')
ax.legend(fontsize=11)
ax.grid(True, linestyle='--', alpha=0.4)

plt.tight_layout()
plt.savefig('plot_4_fitted_curve.pdf', bbox_inches='tight')
plt.show()
```

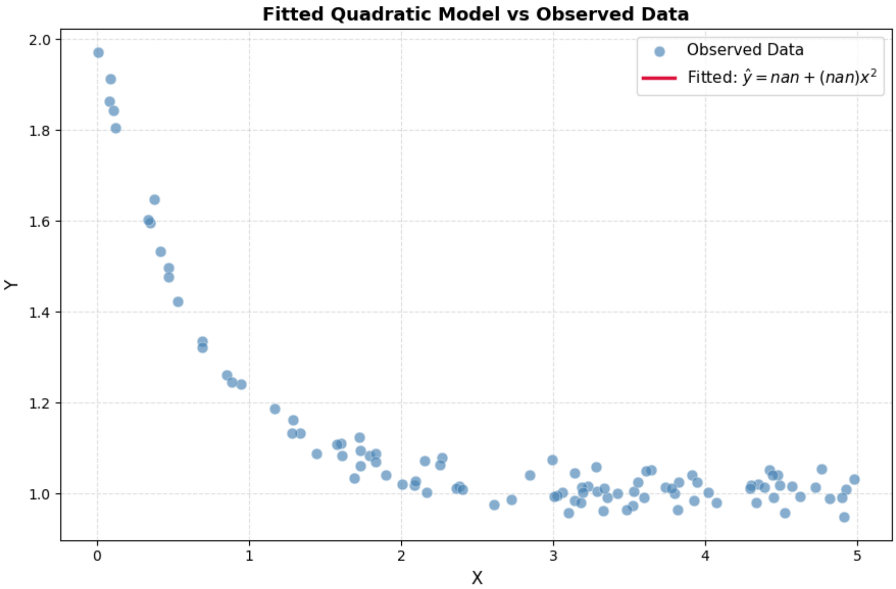


Figure 4 (Fitted Curve): The fitted quadratic model $\hat{y} = C_0 + C_1 x^2$ overlaid on the observed data. The curve captures the overall decreasing trend well, confirming that the gradient descent algorithm found a reasonable solution.

In [7]:

```
# -- Plot 5: Residuals per Observation
    ----------------------------------------
Y_hat_final = C0 + C1 * X2
residuals_final = Y - Y_hat_final
obs_indices = np.arange(1, n + 1)

# Sort by X for visual clarity
sort_idx = np.argsort(X)
X_sorted = X[sort_idx]
res_sorted = residuals_final[sort_idx]
labels_sorted = obs_indices[sort_idx]

fig, ax = plt.subplots(figsize=(14, 5))

colors = ['#e74c3c' if r > 0 else '#2980b9' for r in res_sorted]
ax.vlines(range(n), 0, res_sorted, colors=colors, linewidths=1.2,
    alpha=0.7)
ax.scatter(range(n), res_sorted, color=colors, s=28, zorder=3, alpha
    =0.85)
ax.axhline(0, color='black', linewidth=1.0, linestyle='--')

# Label every 10th observation
```

```python
for i in range(0, n, 10):
    ax.annotate(f'M{labels_sorted[i]}',
                xy=(i, res_sorted[i]),
                xytext=(i, res_sorted[i] + 0.02 * np.sign(res_sorted[i])),
                fontsize=7, ha='center', color='dimgray')

ax.set_xlabel('Observation (sorted by X)', fontsize=12)
ax.set_ylabel('Residual ($y - \\\hat{y}$)', fontsize=12)
ax.set_title('Residuals per Observation (sorted by X)', fontsize=13, fontweight='bold')
ax.grid(True, linestyle='--', alpha=0.35)

from matplotlib.patches import Patch
legend_elements = [Patch(facecolor='#e74c3c', label='Positive residual (under-predicted)'),
                   Patch(facecolor='#2980b9', label='Negative residual (over-predicted)')]
ax.legend(handles=legend_elements, fontsize=10)

plt.tight_layout()
plt.savefig('plot_5_residuals.pdf', bbox_inches='tight')
plt.show()
```

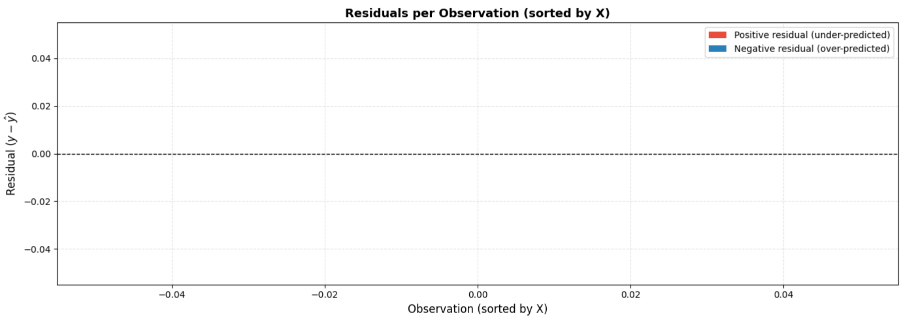


Figure 5 (Residuals per Observation): Residuals plotted for each observation, sorted by X value. Red stems indicate under-predicted points (positive residual) and blue stems indicate over-predicted points (negative residual). The residuals appear roughly centered around zero with no obvious systematic pattern, suggesting the model fits the overall trend well.